

Raka Muhammad Sixmarch

AI & Data Engineering | Data Visualization | Computer Vision | Web Development

Sidoarjo, East Java, Indonesia | +62 813-3293-3065
rakamsixmarch@gmail.com | github.com/raka6march

SUMMARY

Informatics graduate from Telkom University Surabaya, class of 2026, with internship experience in telecommunications data analytics and hands-on projects in machine learning, computer vision, and web development. Experienced in processing large-scale datasets with Python, building Tableau dashboards, and integrating detection models into Flask applications. Academic learning includes big data, machine learning, deep learning, and large language models. Interested in AI and data engineering roles that connect data processing, analysis, and practical application development.

TECHNICAL SKILLS

- **Data Analytics & Engineering:** Data cleaning, exploratory data analysis, aggregation, analytical dataset preparation, SQL, and dashboard development.
- **Programming & Web Development:** Python, SQL, PHP, JavaScript, HTML, CSS, Flask, React, Vite, and Bootstrap.
- **Data Tools & Databases:** Pandas, NumPy, Matplotlib, Tableau, Power BI, MySQL, SQLite, Jupyter Notebook, and Google Colab.
- **Machine Learning & Computer Vision:** Scikit-learn, PyCaret, classification, model evaluation, feature analysis, YOLO11n, YOLOv8, OpenCV, and Roboflow.
- **Development Tools & AI Integration:** Git, GitHub, Gemini API integration, and PDF report generation.

EDUCATION

Bachelor's Degree in Informatics

2022 – 2026

Telkom University Surabaya, Indonesia

- Cumulative GPA: **3.45 / 4.00**.
- Final project: **Football Foul Detection Using YOLO11n with the Roboflow Dataset**.
- Relevant study areas: big data, artificial intelligence, machine learning, deep learning, database systems, and software development.

WORK EXPERIENCE

Data Analytics Intern

Mar 2025 – Apr 2025

PT Telkomsel Regional Surabaya

- Processed more than **10,8 million daily telecommunications records**, as documented in the internship report, using Python and Pandas to prepare monthly site-level analytical datasets.
- Evaluated classification models through PyCaret, including Extra Trees, to classify payload potential into low, medium, and high categories.
- Developed Tableau dashboards presenting site distributions, monthly payload trends, and site-level details to support network performance analysis.
- Configured filters for selected worksheets so users could inspect individual sites while retaining the overall site-class distribution.

SELECTED PROJECTS

Telkomsel Site Potential Analysis

GitHub

Python, Pandas, NumPy, Matplotlib, Tableau

- Extended the original payload-focused internship project to analyze **payload, digital, games, and video** as separate categories.
- Audited source data, aggregated daily metrics by site and month, and assigned relative usage classes using separate tertile thresholds for each category.
- Built dashboard views with category and month filters, site counts, median usage, high-class site percentages, class profiles, and observation coverage.

SELECTED PROJECTS – CONTINUED

FoulVision – Football Foul Review Application

GitHub

Python, Flask, YOLO11n, OpenCV, Roboflow

- Developed a local photo and video review application using YOLO11n to detect three classes: **foul, fouling player, and victim**.
- Implemented JPG/PNG and MP4 uploads, media validation, adjustable confidence thresholds, background processing, and annotated outputs for files up to 200 MB and videos up to five minutes.
- Added slow playback, frame-by-frame video navigation, referee decision notes, and downloadable annotated media, TXT notes, CSV detections, and JSON summaries.
- Designed the workflow to assist human review: video inference operates per frame, while the referee makes the final decision.

Acne Detection with YOLOv8

GitHub

React, Vite, Python, Flask, YOLOv8, MySQL

- Developed an acne and skin-type detection application connecting a React interface with a Flask backend and model inference.
- Implemented user accounts, detection history, and PDF reports, with documented camera-based detection functionality.
- Included optional Gemini and Fonnte integrations for AI analysis and WhatsApp functionality, requiring the respective API configuration.

Employee Permission and Exit/Return Tracking

GitHub

Python, Flask, Jinja, SQLite, WhatsApp Cloud API

- Developed a three-role workflow for employees, supervisors, and administrators, covering permission requests, approval or rejection, and exit/return records.
- Implemented personal QR codes with a two-minute validity period, administrator confirmation, activity history, and CSV exports.
- Added configurable WhatsApp notifications, password hashing, CSRF protection, login attempt limits, and administrative corrections with audit history.

Yakes Medis – Medical Records Application

GitHub

PHP 8, MySQL, Bootstrap, Dompdf

- Developed a portfolio application for patient records, medical history, doctor schedules, clinic visits, medicine catalogs, and PDF reports.
- Added inventory views for stock quantities, minimum stock levels, and expiration dates, alongside operational dashboard summaries.
- Designed relational data structures for clinics, visit queues, vital signs, prescriptions, stock movements, and audit logs, with demonstration data for selected operational modules.

CERTIFICATIONS & TRAINING

- **Gemini Certified Student – University** | Google for Education
Issued Jul 2026; valid until Jul 2029. Foundational knowledge and competencies in using Google AI for education.
- **English Proficiency Test (EPrT)** | Telkom University Language Center
Total score: **450** (Listening: 46; Grammar: 45; Reading: 44). Issued Oct 2025; valid until Oct 2027.
- **Accounting and Management Education** | Pondok Pesantren Wali Barokah, Kediri
Passed the final examination for the June 2022 period.
- **Dakwah Training** | Pondok Pesantren Gadingmangu | Nov 2021
Covered communication ethics, community cultural awareness, organizational knowledge, and national awareness.